Analyzing and Interpreting Motion Graphs: Lab

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Minds On

- Time on x-axis, position on y-axis; right is positive direction
- Watch the walker -> sketch a p-t each round
- “Boards up” on my cue; whisper-check first

Minds-ON

Round 1: Stand at 2 m. Stay still for 5 s

Round 2: Start at 1 m. Walk steadily away for 5 s

Round 3: Start at 6 m. Walk steadily toward 0 for 5 s

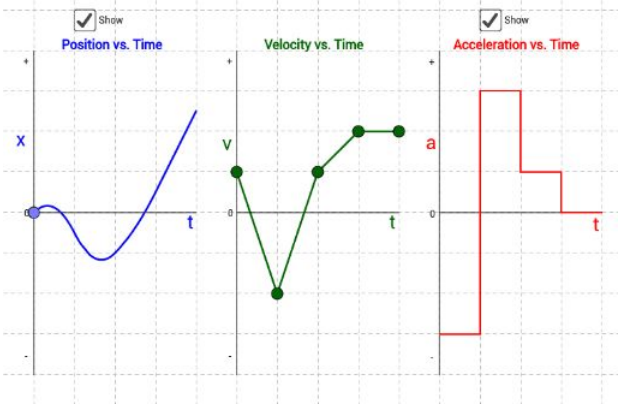
Learning Goal

- ➔ We are learning how to read and draw p-t, v-t, and a-t graphs for 1-D motion
- ➔ We are learning how to convert between these graphs using slope (for velocity and acceleration) and area (for displacement)

Success Criteria

- ➔ I can read p–t, v–t, and a–t graphs and describe the motion with correct units and sign.
- ➔ I can draw the matching graphs (p–t ↔ v–t ↔ a–t) using slope to link them.
- ➔ I can find displacement from a v–t graph (area) and cross-check it on my p–t graph.
- ➔ I can collaborate with my group by doing my role, listening, and helping my group stay on task.

Mini-Review



Big 3:

- p-t slope = velocity
- v-t slope = acceleration
- Area under v-t = displacement

Investigation 1: Toy Car Data collection and Analysis

Objectives

- Collect position-time data for a toy car
- Plot position-time graph and draw line of best-fit
- Calculate velocity from the slope of the line

Materials

- Toy car
- Stopwatch (phones, school laptops or tablets)
- Meter sticks (or measuring device of some sort)
- Masking tape
- Data recording sheet (sheet of paper for recording data)

Data Table and Graph Setup

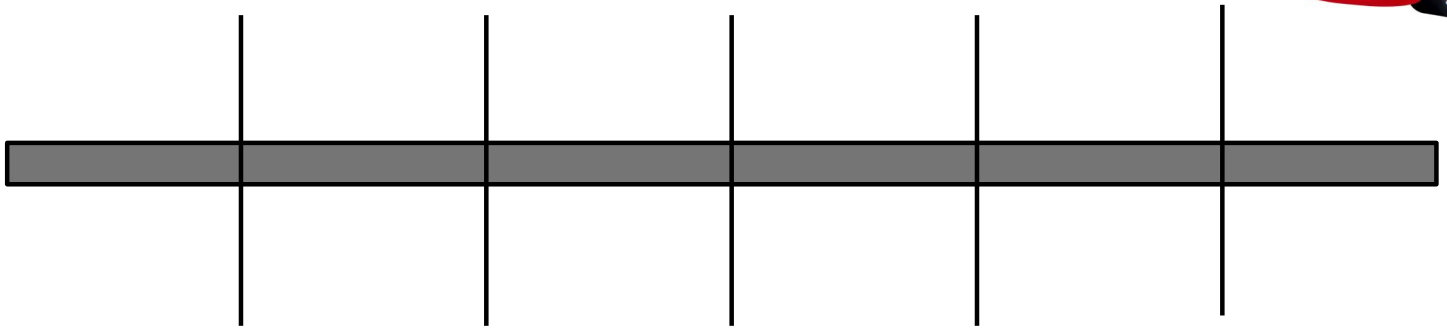
Time (s)	Position (cm)
0	0
2	40
…	…

Graph tips

- Time on horizontal (x) axis, position on vertical (y) axis
- Label axes with units
- Use ruler for straight axes and a colored pen for clarity
- Try to keep the time intervals constant (e.g., every 2 s)

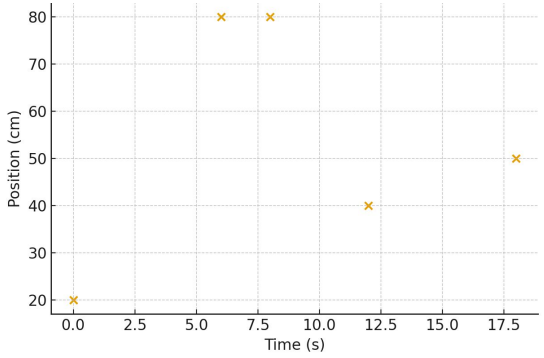
Track Setup and collecting data

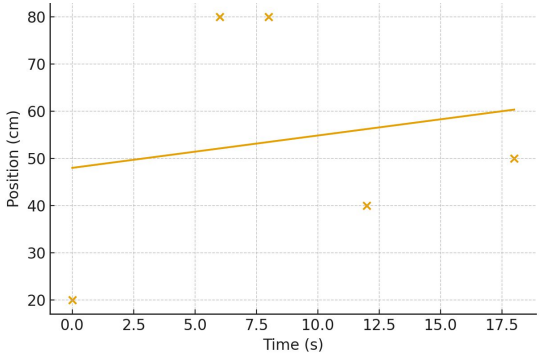
“**20 cm** mark at **2 seconds**
30cm mark at **4 seconds**”



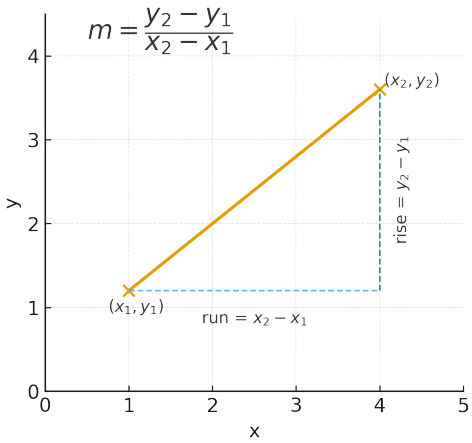
Graph Setup: Example

Time (s)	Position (cm)
0	20
6	80
8	80
12	40
18	50
…	…





Best-Fit Line: Trend Vs Noise



Calculating slope:

1. Pick any two points
2. Plug into formula
3. How can we interpret the slope? What does the slope represent?

Notes

- Do not connect dots
- One straight line that best represents the trend

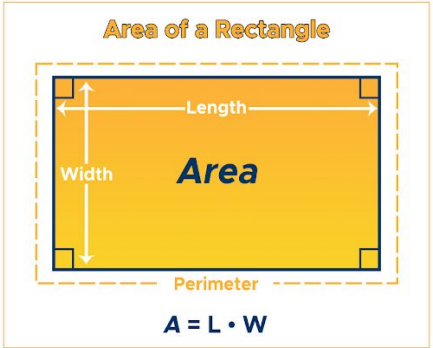
Converting p–t to v–t

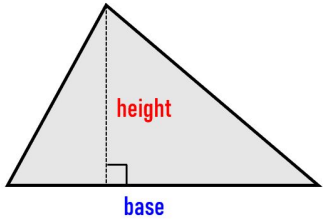
To keep in mind:

- p–t straight line with positive slope → v–t horizontal line above zero
- p–t horizontal line → v–t at zero
- p–t curve getting steeper → v–t increasing

Steps:

1. Break p–t into time segments
2. Compute slope for each segment
3. Plot those slope values as horizontal segments on v–t





$$A = \frac{1}{2}bh$$

b = base

h = height

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Area Under v–t: How to Get Displacement

Displacement = area under the v-t curve

Shapes:

- Rectangle = base × height
- Triangle = ½ × base × height

Example:

- Constant 4m/s for 5s -> 20m
- Acceleration from 0 to 4 m/s over 2s -> 4m

Slope Part 2: Getting Acceleration

Acceleration equals the slope of the v-t graph ($\Delta v/\Delta t$)

To keep in mind:

– Rising v–t → positive a–t

– Flat v–t → zero a–t

– Falling v–t → negative a–t

Steps:

1. Identify intervals on v–t.
2. Compute $a = \Delta v/\Delta t$
3. Plot those values as horizontal segments on a–t.

Track Setup and collecting data (RAMP)

"**20 cm** mark at **2 seconds**
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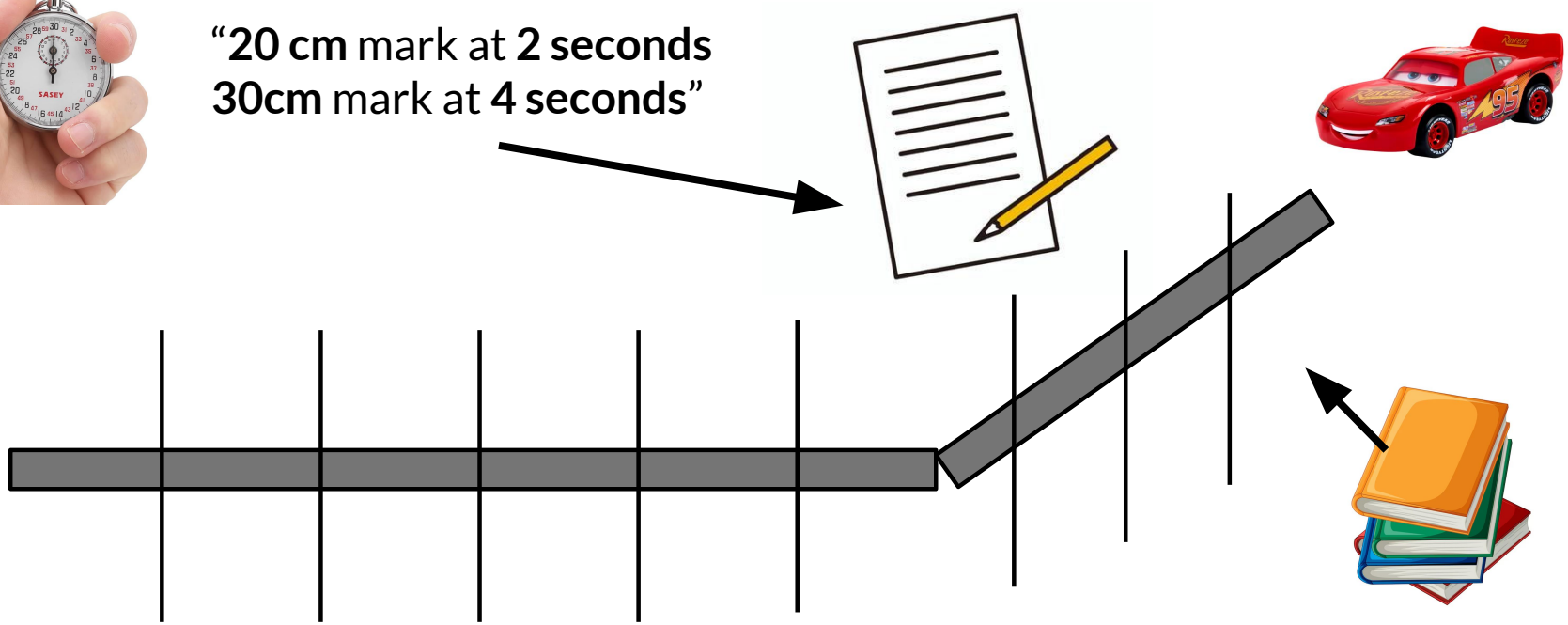

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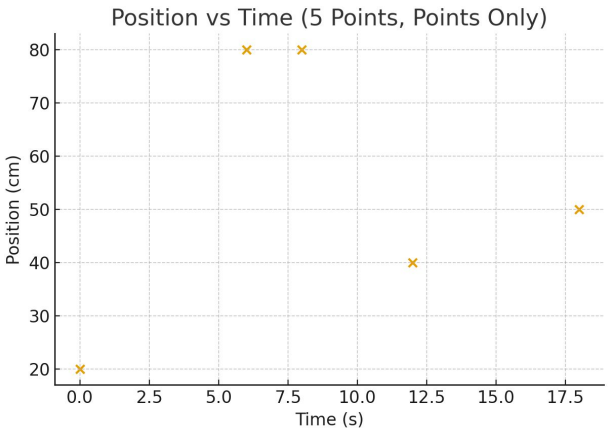
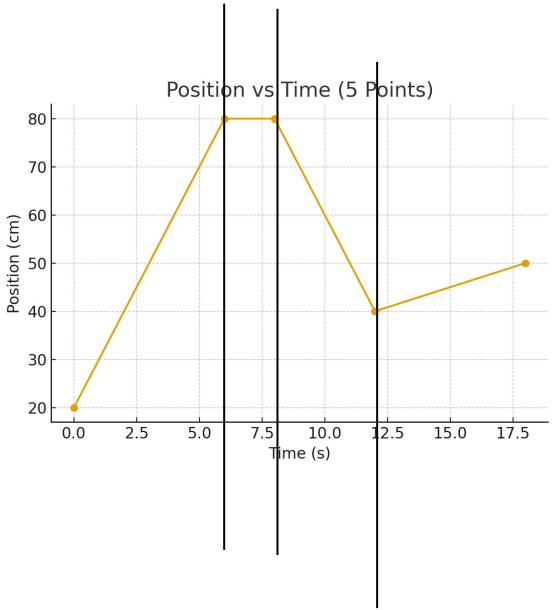


Regions



*****Note, here we will be connecting the points***

Poster Instructions (Ramp)

- Attach your three graphs (glue/tape): p–t (top), v–t (middle), a–t (bottom).
- Label regions on all three: “Region A: ramp” and “Region B: level,” with a vertical dashed transition line
- Show slopes for each region and how you calculated them
- Compute the area under your v–t over a stated time interval and state what it represents (displacement, in metres).
- Write what the slope and area represents in our motion graphs

Gallery Walk: What to Do

Visit two posters (that are not your own)

Sticky note: one accurate connection + one question you may have

Ex: Where is velocity the largest? What p-t feature matches it?

3-2-1 Connect & Reflect

3 connections using slope / area / sign (own data)

2 fixes for a partner (units, sign, axes, slope vs height)

1 question you still have